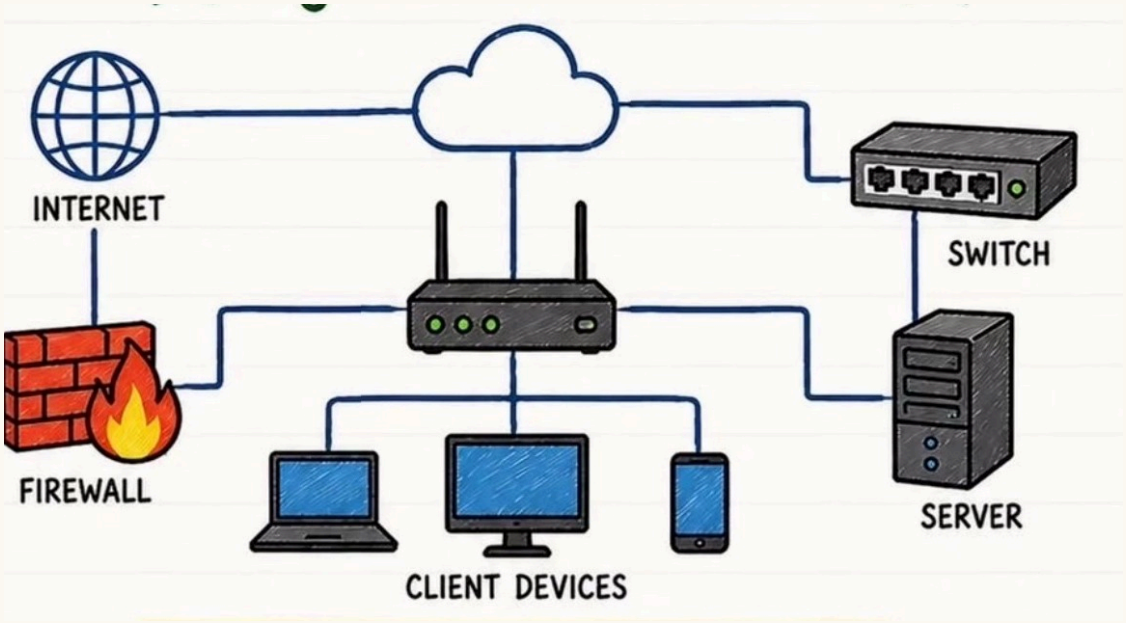
NETWORKING

COMPLETE NOTES

(Beginner to Advanced)



LEARN · UNDERSTAND · APPLY

Notes by @qa_insights · cover artwork by @codyboy_1122 — layout rebuilt as a study script

Contents

1. Introduction to Computer Networking
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11. HTTP & HTTPS
12. Routing, NAT & Gateway
13. Firewalls & Network Security
14. Load Balancer & Proxy
15. Docker Networking
16. Kubernetes Networking
17. Cloud Networking (AWS VPC)
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CHAPTER 01

Introduction to Computer Networking

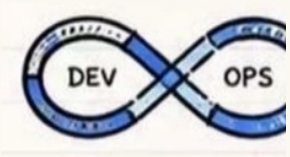
1 What is Networking?



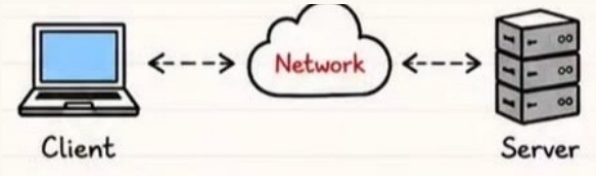
Networking is the process of connecting two or more devices to exchange data and share resources using communication protocols.

2 Why Networking matters in DevOps

- ✔ Enables communication between tools, services & users
- ✔ Critical for CI/CD, monitoring, logging & automation
- ✔ Ensures security, performance, scalability & reliability
- ✔ Foundation of Cloud, Containers & Microservices



3 Client ↔ Server Architecture



- Client (User/Device) sends a request.
- Server processes the request.
- Server sends back the response.

4 LAN, WAN & Internet



LAN (Local Area Network)
Connects devices in a small area like home, office, campus.

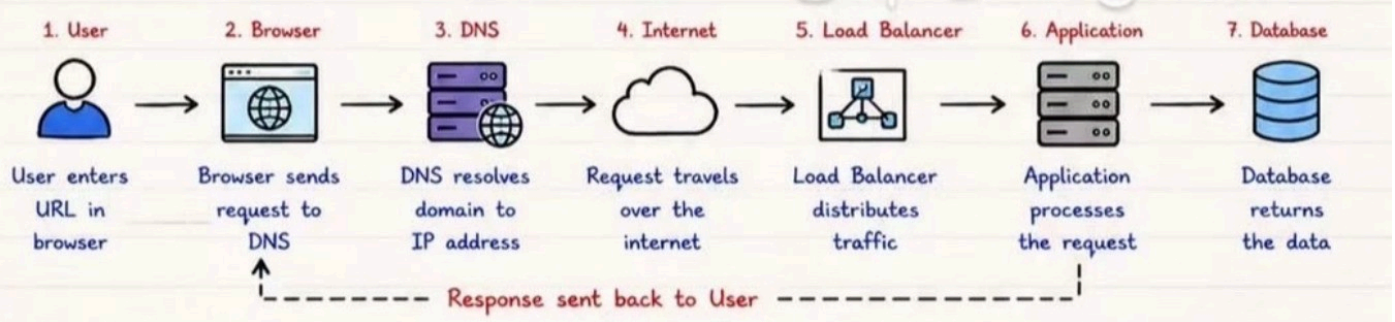


WAN (Wide Area Network)
Connects LANs over large distances like cities, countries.

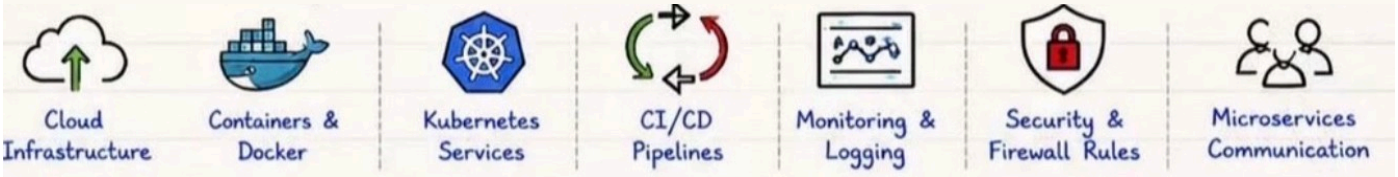


Internet
Global network of networks that connects the world.

5 Real-world Request Flow (Example)



6 Where Networking is used in DevOps



⭐ **Interview Tip**
Networking is one of the most frequently tested topics in DevOps interviews.

💡 **Remember**
Every application request depends on networking before reaching the server.

CHAPTER 02

Network Devices

1 Common Network Devices



Hub

A multi-port repeater. It broadcasts data to all connected devices.



Switch

Connects devices in a LAN. Sends data only to the intended device using MAC addresses.



Router

Connects multiple networks (LAN to WAN). Routes data based on IP addresses.



Gateway

Acts as an entry/exit point between different networks or protocols.



Modem

Modulates and demodulates signals to connect to an ISP over telephone line, cable, or fiber.



Firewall

Monitors and controls incoming and outgoing network traffic based on security rules.

2 Device Comparison Table

Device	Layer (OSI)	Main Function	Works On	Example Use in QA Insights	Key Point
Hub	Layer 1	Broadcasts data to all ports	Bits	Legacy lab environments	No intelligence, causes collisions
Switch	Layer 2	Forwards data to specific device	Frames	QA Insights office LAN	Uses MAC address table
Router	Layer 3	Routes data between different networks	Packets	Connects QA Insights LAN to Internet	Uses IP address & routing table
Gateway	Layer 4-7	Protocol translation between networks	Data / Packets	API Gateway in QA Insights applications	Different protocols can communicate
Modem	Layer 1	Converts digital ↔ analog signals	Bits	QA Insights connects to ISP	Provides Internet access
Firewall	Layer 3-7	Filters & controls traffic for security	Packets / Data	Protects QA Insights servers & network	Allows or blocks based on rules

3 Interview Tip ⭐

Remember the basic rule:

Hub (Dumb) < Switch (Smart) < Router (Smarter) < Firewall (Smartest & Secure)

💡 **Remember**

Devices work together to deliver the right data to the right place securely and efficiently.

CHAPTER 03

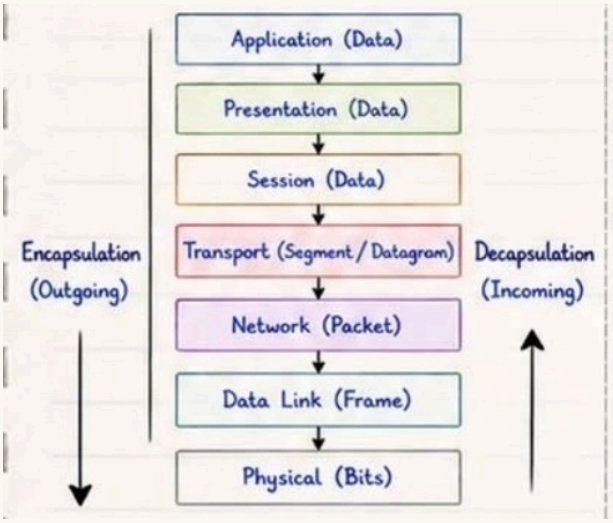
OSI Model (7 Layers)

The OSI (Open Systems Interconnection) model is a conceptual framework that standardizes how data is transmitted over a network.

1 7 Layers Overview

Layer (No.)	Layer Name	Responsibilities	Data Unit (PDU)	Devices at this Layer	Example / Protocols
7	Application Layer	Provides network services to user applications.	Data	—	HTTP, HTTPS, FTP, SMTP, DNS, SSH, DHCP
6	Presentation Layer	Handles data format, encryption, compression, translation.	Data	—	SSL/TLS, JPEG, ASCII, MPEG
5	Session Layer	Manages sessions, establishes, maintains & terminates connections.	Data	—	NetBIOS, RPC, PPTP
4	Transport Layer	Provides end-to-end delivery, segmentation, flow control, error control.	Segment (TCP) Datagram (UDP)	L4 Switch (Load Balancer)	TCP, UDP
3	Network Layer	Logical addressing, routing, path selection, packet forwarding.	Packet	Router (L3)	IP (IPv4, IPv6), ICMP, OSPF, BGP
2	Data Link Layer	Framing, MAC addressing, error detection, access control.	Frame	Switch / Bridge (L2)	Ethernet, PPP, 802.11, MAC
1	Physical Layer	Transmits raw bits over physical medium.	Bits	Hub, Repeater	Cables, Connectors, Hubs, Signal Standards

2 Data Unit (PDU) Flow



> 💡 **Remember**
> OSI model is a reference model. Real networks use TCP/IP model in practice.

3 Easy Memory Trick



- **P** – Physical
- **D** – Data Link
- **N** – Network
- **T** – Transport
- **S** – Session
- **P** – Presentation
- **A** – Application

4 Key Takeaways

- ✓ Upper 3 layers (5–7) focus on application communication.
- ✓ Middle layer (4) ensures reliable end-to-end delivery.
- ✓ Lower 2 layers (1–2) handle actual data transmission.
- ✓ Data flows down from Layer 7 to 1 when sending and up from 1 to 7 when receiving.

> ⭐ **Interview Tip**
> Be ready to explain each layer easily with examples and devices. Remember the memory trick under pressure!

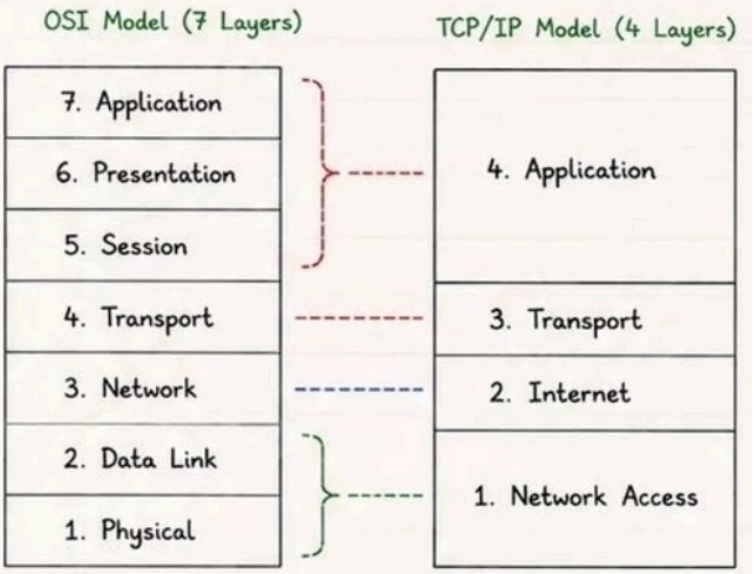
CHAPTER 04

TCP/IP Model (4 Layers)

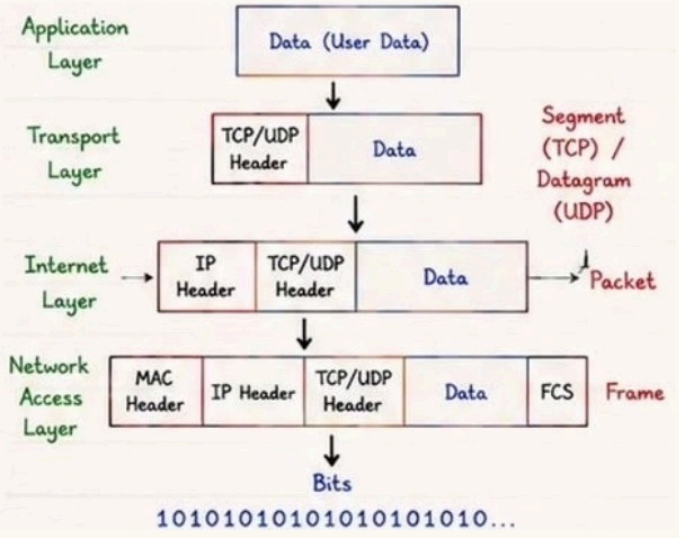
1 4 Layers of TCP/IP Model

Layer (Top to Bottom)	Key Responsibilities	Example Protocols
4 · Application Layer	Provides network services to applications. Handles user interfaces, APIs, data formatting.	HTTP, HTTPS, FTP, DNS, SSH, SMTP, DHCP
3 · Transport Layer	End-to-end communication. Segmentation, reliability, port management.	TCP, UDP
2 · Internet Layer	Logical addressing and routing of packets across networks.	IP (IPv4, IPv6), ICMP, ARP
1 · Network Access Layer	Transmits data on the physical network. Framing, MAC addressing.	Ethernet, Wi-Fi, PPP

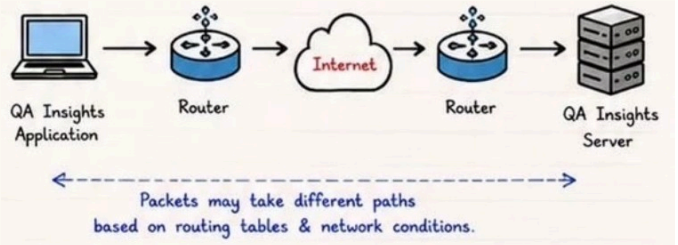
2 OSI vs TCP/IP Model



3 Encapsulation (At Sender)



4 Packet Journey (Sender to Receiver)



> 💡 **Remember**
> Each layer adds its own header (and sometimes trailer) while sending, and removes it while receiving.

5 Why TCP/IP Model is Used?

- ✓ Simple and easy to understand.
- ✓ Widely adopted across the Internet and all modern networks.
- ✓ Supports interoperability between different vendors and systems.
- ✓ Efficient, scalable and works well for large distributed systems.
- ✓ Forms the foundation for protocols used in DevOps tools, Cloud, Containers, and Microservices.

> ⭐ **Interview Tip**
> In interviews, focus on understanding layer functions, real-world examples, and how data moves from one layer to another.

CHAPTER 05

IP Addressing

1 IPv4 vs IPv6

IPv4	IPv6
• 32-bit address • Format: a.b.c.d (0 – 255 each octet) • Example: 192.168.1.10 • Total Addresses: ~4.3 Billion	• 128-bit address • Format: xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx • Example: 2001:0db8:85a3::8a2e:0370:7334 • Total Addresses: ~340 Undecillion

IPv6 was created to overcome IPv4 address exhaustion and provide better scalability & security.

2 Public vs Private IP



Public IP (Internet Reachable)

- Assigned by ISP
- Unique globally
- Used to communicate over the Internet
- Example: 8.8.8.8



Private IP (Internal Use)

- Used inside private networks (LAN/VPC)
- Not routable on the Internet
- Saves public IPs
- Examples: 10.0.0.0/8 · 172.16.0.0/12 · 192.168.0.0/16

3 Static vs Dynamic IP

Static IP

- Manually configured
- Does not change
- Used for servers, devices that need consistent access
- Example: QA Insights Server 49.36.12.5

Dynamic IP

- Assigned automatically by DHCP
- Can change over time
- Used for client devices
- Example: QA Insights Developer Laptop (Changes periodically)

4 Loopback Address

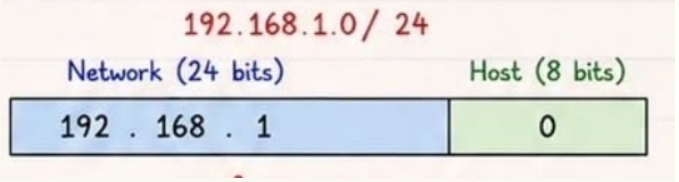


- Used by a device to communicate with itself.
- Used for testing and local development.
- IPv4 Loopback: 127.0.0.1
- IPv6 Loopback: ::1

Packets sent to loopback never leave the device.

5 CIDR (Classless Inter-Domain Routing) – Introduction

- CIDR is a method of allocating IP addresses and routing IP packets.
- It replaces traditional classful addressing.
- Notation: IP Address / Prefix Length
- Example: 192.168.1.0/24
  - /24 means first 24 bits are network part
  - Remaining 8 bits are for hosts
- CIDR helps in efficient IP usage and route summarization.



Supports $2^8 - 2 = 254$ usable hosts

Common CIDR Ranges			
	/8	→	16,777,214 hosts
	/16	→	65,534 hosts
	/24	→	254 hosts
	/30	→	2 hosts (Point-to-Point)

⭐ **Interview Tip**

Understand IP types, ranges, and CIDR deeply – these are frequently asked in DevOps & Networking interviews!

💡 **Remember**

IP addressing is the foundation of all networking. Without IP, devices cannot communicate!

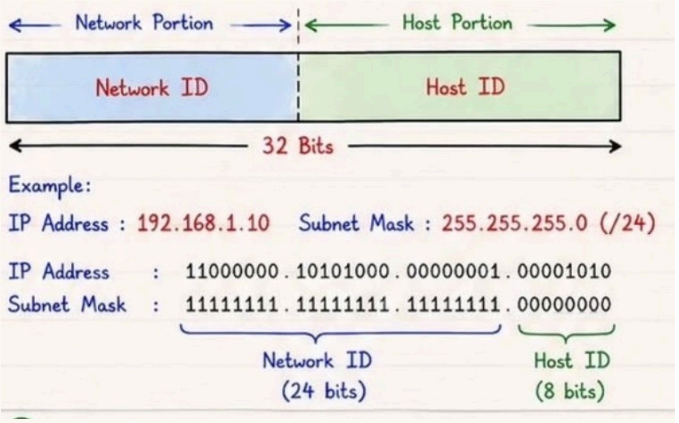
CHAPTER 06

Subnetting & CIDR

1 Key Terms

- **Network ID** : Identifies the network portion of an IP.
- **Host ID** : Identifies the specific device in that network.
- **Subnet Mask** : Divides IP into network and host portions.
- **CIDR Notation** : Short and efficient way to represent IP and subnet mask.
- **Available Hosts** : Total usable IP addresses in a subnet.

2 IP Address Structure (IPv4)



3 Subnet Mask

A subnet mask is a 32-bit number that tells which part of the IP is **Network ID** and which part is **Host ID**.

Common Subnet Masks

CIDR	Subnet Mask	Network / Host Bits
/8	255.0.0.0	8 / 24
/16	255.255.0.0	16 / 16
/24	255.255.255.0	24 / 8
/25	255.255.255.128	25 / 7
/26	255.255.255.192	26 / 6
/27	255.255.255.224	27 / 5
/28	255.255.255.240	28 / 4
/30	255.255.255.252	30 / 2

4 CIDR Notation

CIDR (Classless Inter-Domain Routing) removes the need for classful networks and allows variable length subnetting.

Format: **IP Address / Prefix Length**

Examples

- **192.168.1.0/24** → First 24 bits are network portion
- **10.0.0.0/8** → First 8 bits are network portion
- **172.16.5.0/16** → First 16 bits are network portion

⭐ **Interview Tip**

CIDR helps in efficient IP allocation, route summarization and reduces routing table size in Routers.

5 Available Hosts Calculation

Formula

Total Hosts = 2^n

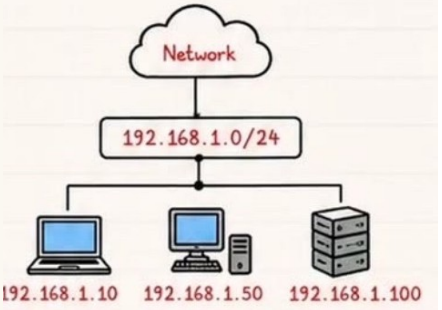
Usable Hosts = $2^n - 2$

Where **n** = number of host bits
(We subtract 2 for Network ID and Broadcast ID)

Example Calculation

Given IP: **192.168.1.10/24**
Subnet Mask: **255.255.255.0**

CIDR	/24
Host Bits (n)	8
Total Hosts (2^n)	2^8 = 256
Usable Hosts ($2^n - 2$)	256 − 2 = 254
Host Range	192.168.1.1 – 192.168.1.254
Network ID	192.168.1.0
Broadcast ID	192.168.1.255



💡 **Remember**

- ✓ Network ID → First IP in the subnet (all host bits = 0)
- ✓ Broadcast ID → Last IP in the subnet (all host bits = 1)

CHAPTER 07

MAC Address & ARP

1 MAC Address (Media Access Control)

- A unique hardware address assigned to Network Interface Card (NIC).
- Works at **Data Link Layer (Layer 2)**.
- **48-bit address** (6 groups of 2 hex digits).
- Format: **00:1A:2B:3C:4D:5E**
- Uniquely identifies a device on a local network.



2 Physical vs Logical Address

Physical Address (MAC)	Logical Address (IP)
• Hardware address • Layer 2 (Data Link) • Assigned by Manufacturer • Used for local network communication • Example: 00:1A:2B:3C:4D:5E	• Software address • Layer 3 (Network) • Assigned manually or by DHCP • Used for end-to-end communication • Example: 192.168.1.10

3 ARP (Address Resolution Protocol)

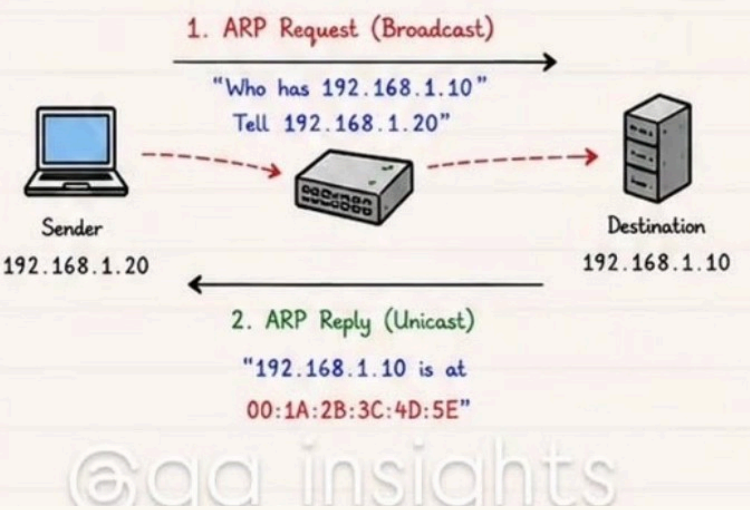
- ARP maps **IP address** (Logical) to **MAC address** (Physical).
- Works within the local network.
- If sender doesn't know the MAC of destination IP, it sends an **ARP Request (Broadcast)**.
- Destination replies with **ARP Reply (Unicast)** containing its MAC.



ARP helps find the MAC address of a device using its IP.

> 💡 **Remember**
> ARP is like asking: "Who has **192.168.1.10**? Tell **192.168.1.20** that I am **192.168.1.10**."

4 ARP Process (Broadcast & Reply)



5 Switch MAC Table

- Switch learns MAC addresses from incoming frames.
- Stores in MAC Address Table with port information.
- Used to forward frames only to the correct port (not broadcast).

Example MAC Table

MAC Address	Port
00:1A:2B:3C:4D:5E	Port 1
00:1A:2B:3C:4D:6F	Port 2
00:1A:2B:3C:4D:7A	Port 3
00:1A:2B:3C:4D:8B	Port 4
...	...

6 Packet Delivery Flow



1. Sender checks if destination IP is in same subnet.
2. Uses ARP to get destination MAC.
3. Sends frame to Switch (with destination MAC).
4. Switch looks up MAC Table and forwards to correct port.
5. Destination receives the frame and processes it.

> ✅ **Key Points**
> - ✓ MAC = Hardware Address (Layer 2)
> - ✓ IP = Logical Address (Layer 3)
> - ✓ ARP maps IP → MAC
> - ✓ Switch uses MAC Table for intelligent forwarding

> ⭐ **Interview Tip**
> Be clear about the difference between MAC & IP, how ARP works, and how switches learn & forward frames. These are commonly asked in DevOps, Network & Cloud interviews.

CHAPTER 08

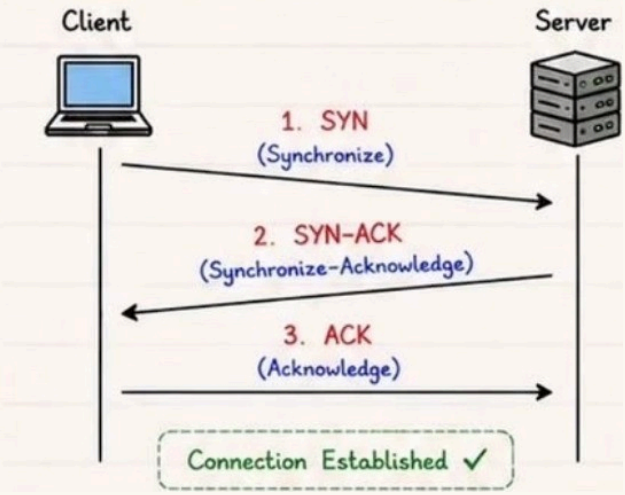
TCP vs UDP

TCP and UDP are Transport Layer protocols used to transmit data between applications over an IP network.

1 Connection-Oriented vs Connectionless

- **TCP is Connection-Oriented**
  A connection is established before data transfer and closed after completion.
- **UDP is Connectionless**
  No connection setup. Data is sent directly without establishing a session.

2 Three-Way Handshake (TCP)



3 Reliability vs Speed

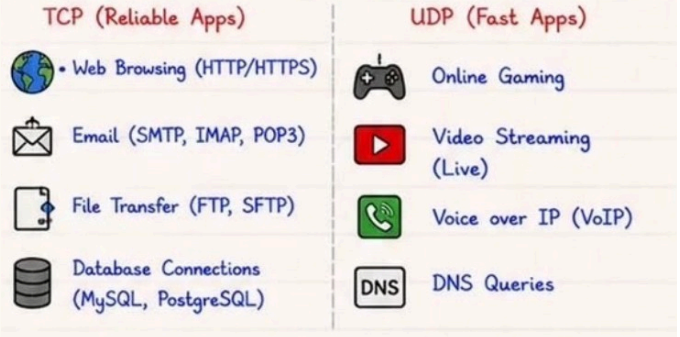
TCP

- ✓ Reliable delivery
- ✓ Ensures data arrives in order
- ✓ Error checking & retransmission
- ✓ Slower due to overhead

UDP

- ✓ Best-effort delivery
- ✓ No guarantee of order or delivery
- ✓ No retransmission or error recovery
- ✓ Faster and lightweight

4 Real-World Examples



5 TCP vs UDP Comparison

Feature	TCP	UDP
Type	Connection-Oriented	Connectionless
Connection Setup	Required (Three-way Handshake)	Not Required
Reliability	Reliable (ACK + Retransmission)	Best-effort (No guarantees)
Ordering	Maintains Order	No Order Guarantee
Error Checking	Yes (Checksum, ACK)	Yes (Checksum only)
Data Delivery	Guaranteed	Not Guaranteed
Speed	Slower	Faster
Overhead	Higher (Header + Handshake)	Lower (Minimal Header)
Data Type	Stream (Continuous)	Datagram (Independent packets)
Header Size	20-60 Bytes	8 Bytes
Use Cases	Web, Email, File Transfer, DB, SSH	Gaming, Streaming, DNS, VoIP, IoT

⭐ **Interview Tip**

If data loss is acceptable but speed is critical → use **UDP**.

If accuracy and reliability are important → use **TCP**.

💡 **Remember**

TCP = **T**rust, **C**onfirm, **P**rotect

UDP = **U**ser, **D**atagram, **P**erformance

💡 **Key Takeaway**

TCP is like a **phone call** (connection, reliable, ordered).

UDP is like a **walkie-talkie** (no setup, fast, may lose packets).



CHAPTER 09

Ports & Common Protocols

Ports are logical endpoints used by protocols to identify applications and services on a host.

Protocol / Service	Port	Transport	Purpose
HTTP (HyperText Transfer Protocol)	80	TCP	Unencrypted web traffic
HTTPS (HTTP Secure)	443	TCP	Encrypted web traffic (SSL/TLS)
SSH (Secure Shell)	22	TCP	Secure remote login & command execution
DNS (Domain Name System)	53	UDP/TCP	Domain name resolution
FTP (File Transfer Protocol)	21 (Control) 20 (Data)	TCP	File transfer (unencrypted)
SFTP (SSH File Transfer Protocol)	22	TCP	Secure file transfer over SSH
SMTP (Simple Mail Transfer Protocol)	25	TCP	Sending emails
Database Ports			
MySQL	3306	TCP	MySQL Database
PostgreSQL	5432	TCP	PostgreSQL Database
MongoDB	27017	TCP	MongoDB Database
Redis	6379	TCP	Redis Cache / Key-Value Store

✅ Key Points

- Port + IP address uniquely identifies a service.
- Well-known ports : 0 – 1023
- Registered ports : 1024 – 49151
- Dynamic / Private ports : 49152 – 65535

⭐ Interview Tip

Always mention the protocol along with the port (e.g., 80/TCP, 53/UDP).

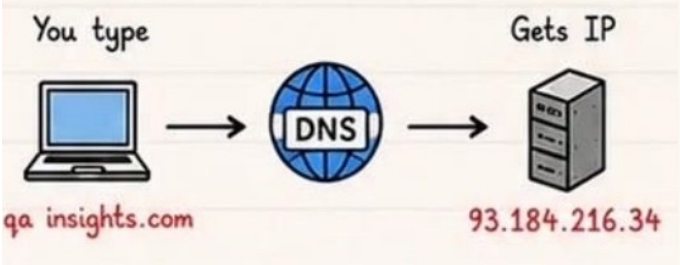
Firewall rules are based on IP, port and protocol.

CHAPTER 10

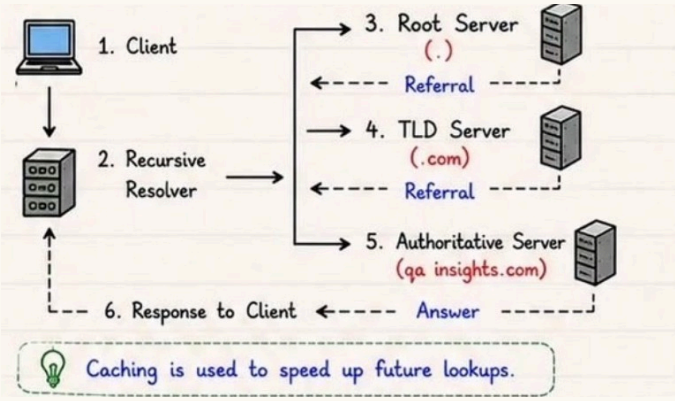
DNS (Domain Name System)

1 What is DNS?

DNS is the Internet's phonebook. It translates human-readable domain names into IP addresses that computers use to communicate.



2 DNS Resolution Flow

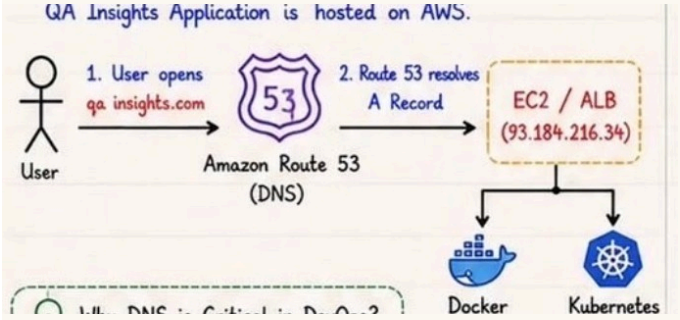


3 Important DNS Record Types

Record Type	Purpose	Example	Explanation
A A Record (Address)	Maps a domain name to an IPv4 address.	qa-insights.com → 93.184.216.34	Points the domain to an IPv4 address.
CNAME CNAME Record (Canonical Name)	Maps one domain name to another domain name.	www.qa-insights.com → qa-insights.com	Alias of another domain. No IP address.
MX Record (Mail Exchange)	Specifies mail server for the domain.	qa-insights.com → mail.qa-insights.com (Priority 10)	Tells where to deliver email for the domain.
TXT TXT Record (Text)	Stores text information about the domain.	qa-insights.com → "v=spf1 include:_spf.qa-insights.com ~all"	Used for SPF, ownership verification, policies, etc.

4 DevOps Example

QA Insights Application is hosted on AWS.



Why DNS is Critical in DevOps?

- ✓ Maps domains to load balancers / services
- ✓ Enables blue-green / canary deployments
- ✓ Works with CDNs, monitoring, email, etc.

5 Quick Reference (Common DNS Record Summary)

Type	Use Case
A	Domain → IPv4 address
AAAA	Domain → IPv6 address
CNAME	Domain alias (one name to another)
MX	Mail server configuration
TXT	SPF, DKIM, verification, policies
NS	Authoritative name servers
PTR	Reverse DNS lookup (IP → Domain)

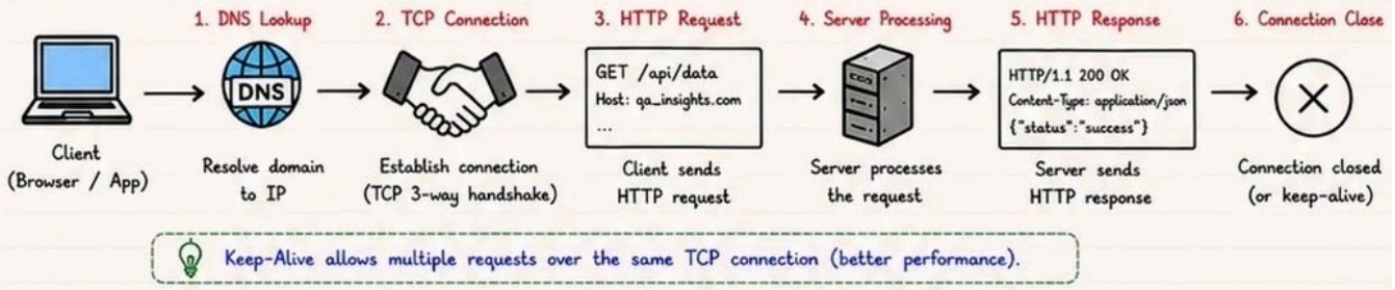
Interview Tip

Always understand how DNS works, record types, and how caching affects resolution time.

CHAPTER 11

HTTP & HTTPS

1 Request Lifecycle



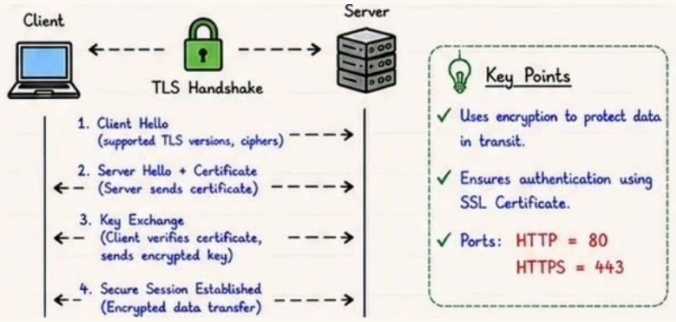
2 HTTP Methods

Method	Purpose	Idempotent	Safe
GET	Retrieve data	✓	✓
POST	Create new resource	×	×
PUT	Update entire resource	✓	×
PATCH	Update partial resource	×	×
DELETE	Delete resource	✓	×
HEAD	Get headers only	✓	✓
OPTIONS	Get supported methods	✓	✓

3 HTTP Status Codes (Most Important)

1xx	Informational	Request received, continuing process
2xx	Success	200 OK, 201 Created, 204 No Content
3xx	Redirection	301 Moved Permanently, 302 Found
4xx	Client Error	400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 429 Too Many Requests
5xx	Server Error	500 Internal Server Error, 502 Bad Gateway, 503 Service Unavailable, 504 Gateway Timeout

4 SSL/TLS (How HTTPS Works)



5 REST APIs

- REST (Representational State Transfer) is an architectural style for APIs.
- Uses HTTP methods to perform operations on resources.
- Resources are identified using URLs.
- Common data format: **JSON / XML**.

Example

```
GET    /api/users      → Get all users
POST   /api/users      → Create user
GET    /api/users/123  → Get user by ID
PUT    /api/users/123  → Update user
DELETE /api/users/123  → Delete user
```

6 HTTPS Benefits



- ✓ Encrypts data → prevents eavesdropping.
- ✓ Authenticates server → prevents man-in-the-middle attacks.
- ✓ Ensures data integrity → data cannot be tampered.
- ✓ Improves user trust (padlock icon in browser).
- ✓ Required for modern web apps, APIs, payments, logins.

7 Quick Reference

Protocol	HTTP (80) / HTTPS (443)
Purpose	Web communication
Data	HTTP → Plain Text HTTPS → Encrypted (TLS)
Security	HTTP → No Security HTTPS → Secure
Best Practice	Always use HTTPS in production

⭐ **Interview Tip**

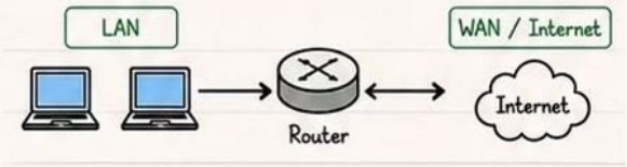
Explain the request lifecycle, difference between HTTP & HTTPS, and when to use each HTTP method.

CHAPTER 12

Routing, NAT & Gateway

1 Router (What it does)

- A router connects multiple networks and forwards packets based on IP addresses.
- It uses a routing table to decide the best path.



2 Routing (How packets find the path)

Routing Table (Example)

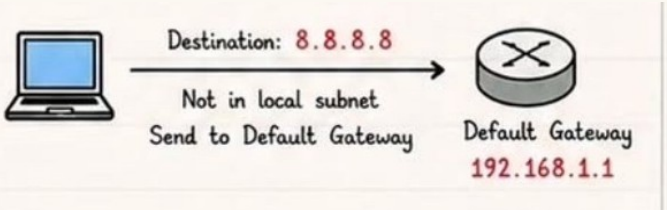
Destination Network	Next Hop
192.168.1.0/24	Directly Connected
10.0.0.0/24	10.0.0.1
0.0.0.0/0	203.0.113.1

💡 **Key Point**

0.0.0.0/0 is the **Default Route**. Used for all unknown destinations (Internet).

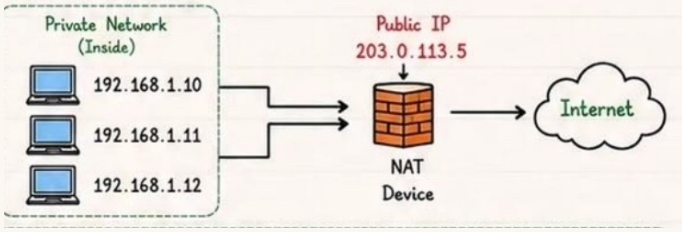
3 Default Gateway

- A default gateway is the router's IP address in your local network.
- If the destination IP is outside the local subnet, traffic is sent to the default gateway.



4 NAT (Network Address Translation)

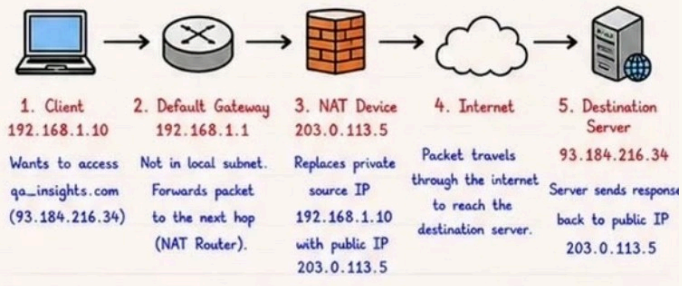
- NAT allows multiple private IPs to share one public IP address.
- Hides internal network from the internet.



Types: SNAT (Source NAT) → for outgoing traffic

DNAT (Destination NAT / Port Forwarding) → for incoming traffic

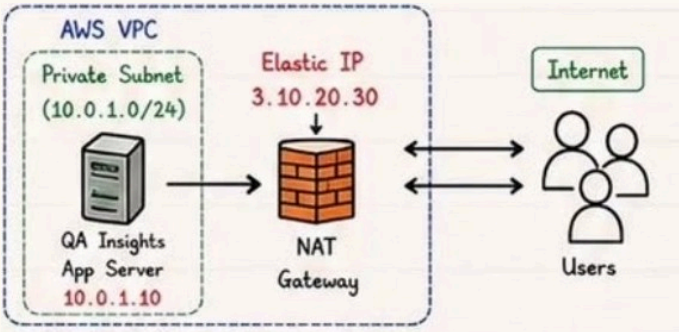
5 Packet Journey (Inside → Internet)



Return Path: Server → Internet → NAT (203.0.113.5) → 192.168.1.10 (Original Client)

6 Real-World Example

QA Insights Application is hosted inside AWS VPC and accessed by users over the Internet.



- Users access https://qa_insights.com
- DNS resolves to 3.10.20.30 (Elastic IP).
- Traffic goes to NAT Gateway → routed to App Server (10.0.1.10).
- Response follows the reverse path.

⭐ **Interview Tip**

- ✓ Router decides the path (Routing).
- ✓ Default Gateway is your exit door to other networks.
- ✓ NAT hides private IPs and enables internet access.
- ✓ Understand packet flow end-to-end.

🔧 **Common Commands**

```
Show Route Table      : ip route  (Linux)
Check Default Gateway : ip route | grep default
Test Connectivity     : ping 8.8.8.8
Trace Route           : traceroute qa-insights.com
```

💡 **Remember**

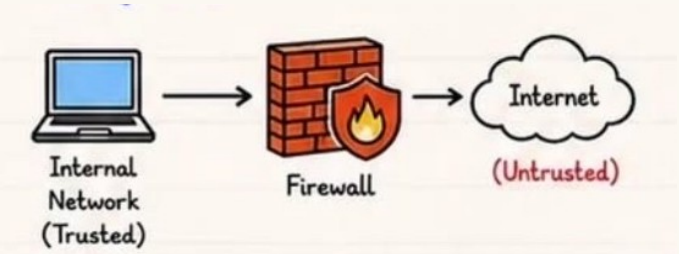
- ✓ Private IPs: 10.x.x.x, 172.16–31.x.x, 192.168.x.x
- ✓ Default Gateway = Router IP in your subnet
- ✓ NAT ≠ Routing (NAT translates, Router routes)

CHAPTER 13

Firewalls & Network Security

1 What is a Firewall?

A firewall monitors and controls incoming and outgoing network traffic based on security rules.



Firewalls act as a barrier between trusted internal networks and untrusted external networks.

2 Stateful vs Stateless Firewall

Stateless Firewall	Stateful Firewall
Examines each packet independently.	Tracks the state of active connections.
Works at L3/L4 (IP, Port, Protocol).	Works at L3/L4 and keeps connection state table.
Faster but less secure.	Slightly slower but more secure.
Allows/blocks based only on rules.	Allows return traffic automatically for valid connections.
Example: Basic ACLs, NACLs.	Example: Most modern firewalls, AWS Security Groups.

3 Security Groups (Stateful)

- Virtual firewall for instances.
- Stateful: Return traffic is automatically allowed.
- Rules are allow only.
- Operates at instance level.
- Example: AWS Security Group, Azure NSG (stateful).

Example (Allow Rules)

Type	Protocol	Port	Source
SSH	TCP	22	203.0.113.10/32
HTTP	TCP	80	0.0.0.0/0
HTTPS	TCP	443	0.0.0.0/0

✓ Only defined traffic is allowed. Everything else is denied.

4 Network ACL (NACL) (Stateless)

- Operates at subnet level (not instance).
- Stateless: You must allow both inbound and outbound.
- Rules are allow and deny.
- Evaluated in order (lowest rule number first).
- Example: AWS NACL, Azure NSG (stateless mode).

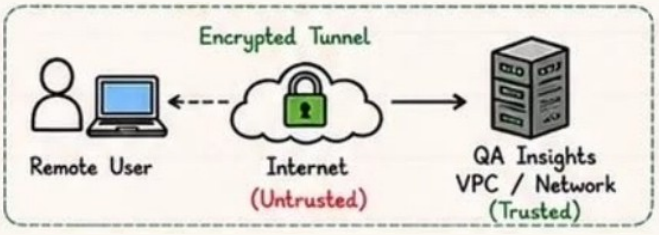
Example (Inbound Rules)

Rule #	Type	Protocol	Port	Source	Action
100	HTTP	TCP	80	0.0.0.0/0	ALLOW
110	SSH	TCP	22	203.0.113.10/32	ALLOW
120	ALL	ALL	ALL	0.0.0.0/0	DENY

✓ If a packet doesn't match any ALLOW rule, it's DENIED.

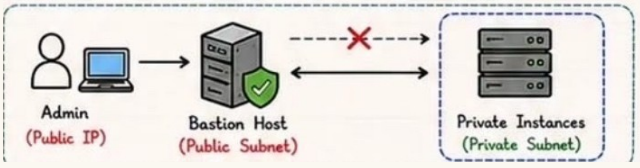
5 VPN (Virtual Private Network)

- Creates an encrypted tunnel over the internet.
- Used for secure remote access or site-to-site connectivity.
- Types: Site-to-Site VPN, Client-to-Site VPN.
- Protocols: IPsec, OpenVPN, SSL/TLS.



6 Bastion Host (Jump Server)

- A secure gateway host to access private servers.
- Direct access to private instances is blocked.
- Admins SSH/RDP to Bastion first, then to internal servers.
- Best practice: Enable MFA, logging, least privilege access.



⭐ Key Points

- ✓ Use least privilege – allow only required traffic.
- ✓ Prefer stateful firewalls (SG) for simplicity.
- ✓ NACLs are for subnet-level control & extra layer.
- ✓ Always encrypt traffic using VPN/TLS.
- ✓ Use Bastion Host – never expose private servers.

💡 Interview Tip

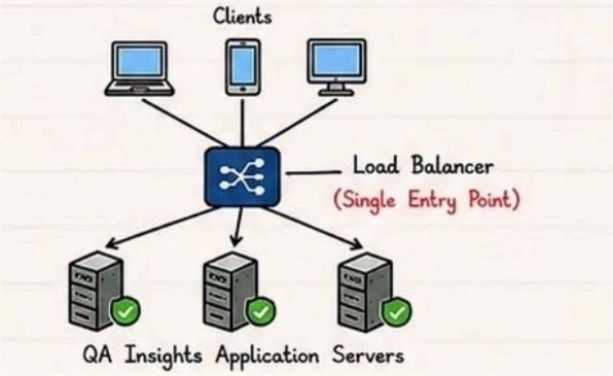
- Stateful firewall remembers connections, stateless does not.
- Security Groups = stateful, NACL = stateless.
- Default behavior: SG = Deny all inbound, Allow all outbound. NACL = Allow all for both (you must change range).
- VPN and Bastion Host improve security posture in cloud.

CHAPTER 14

Load Balancer & Proxy

1 Why Load Balancer?

- Distributes incoming traffic across multiple servers.
- Improves availability, scalability and performance.
- Prevents single point of failure.
- Performs health checks and routes only to healthy servers.



Key Point

Load balancer hides backend servers, provides high availability and seamless scaling.

2 L4 vs L7 Load Balancing

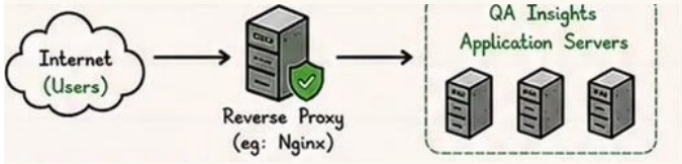
Feature	L4 (Transport Layer)	L7 (Application Layer)
Layer	OSI Layer 4 (TCP/UDP)	OSI Layer 7 (HTTP/HTTPS)
Basis	IP, Port, Protocol	Content (URL, Host, Header)
Speed	Very Fast	Slightly Slower
Example Use	Databases, TCP apps, Microservices	Web apps, APIs, SSL offloading, Content routing
Example Tools	AWS NLB, HAProxy (TCP)	AWS ALB, Nginx, Envoy

3 Forward Proxy (Client-side Proxy)



- Acts on behalf of clients to access the internet.
- Hides internal IPs from external websites.
- Used for caching, filtering, access control.

4 Reverse Proxy (Server-side Proxy)



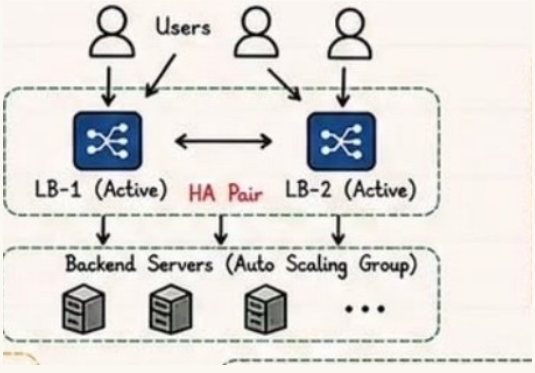
- Sits in front of backend servers.
- Receives client requests, forwards to appropriate server.
- Provides SSL termination, caching, compression, security.
- Hides backend servers from clients.

5 Nginx (Popular Reverse Proxy / Load Balancer)

- ✓ High performance, lightweight and open-source.
- ✓ Works as Web Server, Reverse Proxy, Load Balancer, API Gateway.
- ✓ Supports L4 & L7 load balancing.
- ✓ SSL termination, caching, compression.
- ✓ Widely used in DevOps, Microservices, Kubernetes Ingress Controllers.

6 High Availability (HA) Concepts

- Multiple load balancers (Active-Active)
- Health checks remove unhealthy nodes.
- Auto scaling adds/removes servers.
- Use DNS or Anycast IP for LB redundancy.
- Backups, monitoring and alerts are must.



Remember

- ✓ HA = No Single Point of Failure
- ✓ Use multiple components at every layer (LB, App, DB, Network).
- ✓ Design for failure, not just for performance.

Interview Tip

Know the difference between L4 & L7. Understand when to use Proxy vs Load Balancer. Be ready with examples: Nginx, HAProxy, AWS ALB/NLB.

Common Tools

Load Balancers: Nginx, HAProxy, AWS ALB/NLB, GCP Cloud Load Balancer, Azure Load Balancer

Reverse Proxy: Nginx, Traefik, Envoy

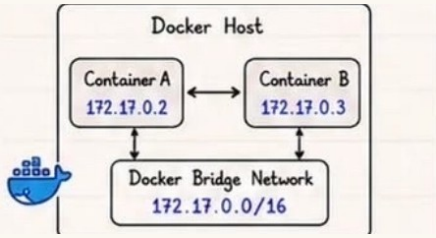
Forward Proxy: Squid, Nginx

CHAPTER 15

Docker Networking

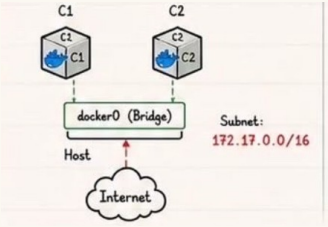
1 Docker Network Overview

- Docker provides multiple network drivers to enable containers to communicate securely and efficiently.
- Use `docker network ls` to list networks.



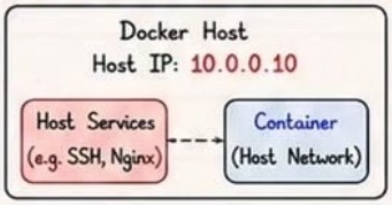
2 Bridge Network (Default)

- Default network mode.
- Containers connect via a virtual bridge (docker0) on the host.
- Containers can talk to each other and access external networks through NAT.



3 Host Network

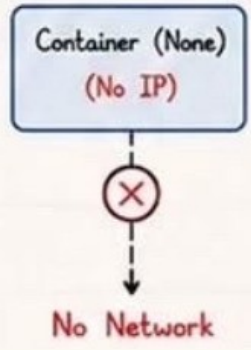
- Container shares the host's network namespace.
- No isolation – container gets host's IP.
- High performance, low latency.



💡 **Use Case:**
High performance apps, monitoring, network tools.

4 None Network

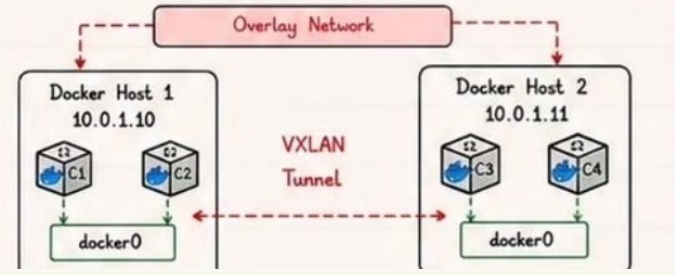
- Container has no networking.
- No IP address, no routing.
- Complete isolation.





5 Overlay Network

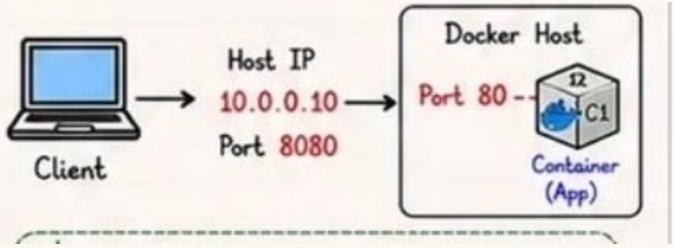
- Used for multi-host container communication.
- Creates a virtual network across Docker hosts.
- Uses VXLAN under the hood.



💡 **Use Case:**
Swarm mode, Kubernetes, multi-host microservices.

6 Port Mapping (Publish Ports)

- Exposes container ports to the host and external world.
- Format: `-p <host_port>:<container_port>`



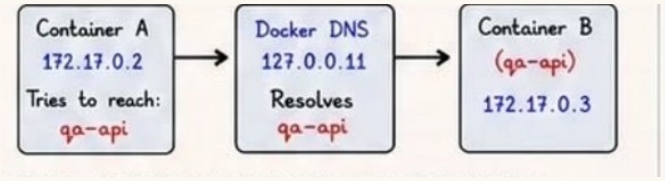
💡 **Example:**

```
docker run -d -p 8080:80 nginx
```

Access Nginx on http://10.0.0.10:8080

7 Docker DNS (Embedded DNS)

- Docker provides an internal DNS server (127.0.0.11) for container name resolution.
- Containers can resolve other containers by name.



💡 **Example:**
Container A calls http://qa-api:8080
✓ DNS resolves "qa-api" to 172.17.0.3

⭐ Key Points

- ✔ **Bridge** = Default network (isolated but connected)
- ✔ **Host** = Best performance
- ✔ **None** = No network access
- ✔ **Overlay** = Multi-host communication
- ✔ **Port Mapping** = Expose services
- ✔ **Docker DNS** = Service discovery by name

⭐ Interview Tip

Understand when to use each network mode. Know port mapping, container name resolution, and overlay for multi-host setups.

🔧 Common Commands

```
docker network ls
docker network inspect <network>
docker run -d --network
<network_name> nginx
docker run -d -p 8080:80 nginx
```

💡 Remember

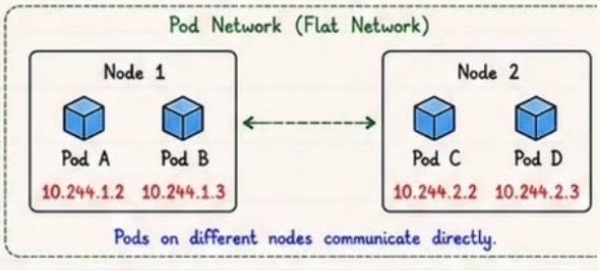
- ✔ Containers on same network communicate by IP or Name.
- ✔ Port mapping exposes container ports.
- ✔ Overlay enables containers on different hosts to talk.

CHAPTER 16

Kubernetes Networking

1 Pod Network

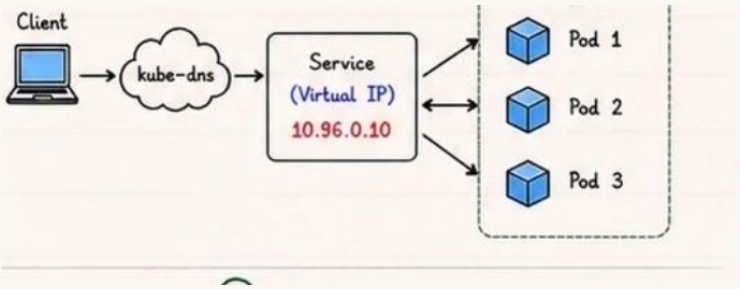
- Each Pod gets its own IP address.
- Pods can communicate with each other across nodes without NAT.
- Implemented using CNI plugins (Flannel, Calico, Cilium, Weave).



> 💡 **Key Point**
> Every Pod IP is reachable from every other Pod in the cluster.

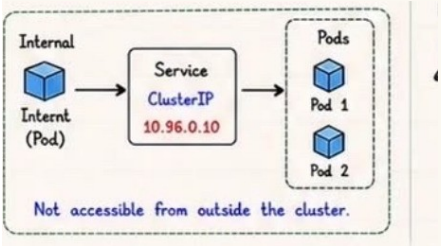
2 Service (Overview)

- Service provides a stable endpoint (IP + DNS) to a set of Pods.
- Decouples clients from Pod lifecycle.
- Kubernetes Service Types: **ClusterIP | NodePort | LoadBalancer | Ingress**



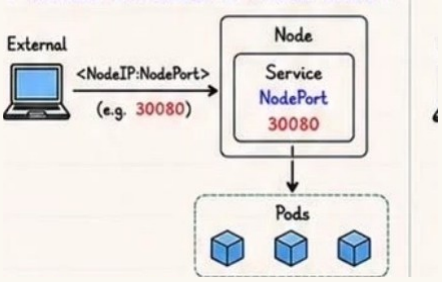
3 ClusterIP (Default)

- Exposes the Service on a cluster-internal IP.
- Accessible only inside the cluster.
- Used for internal communication.



4 NodePort

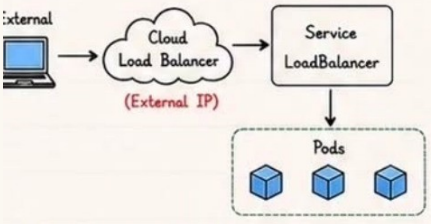
- Exposes the Service on each Node's IP at a static port (30000–32767).
- Accessible from outside via **<NodeIP>:<NodePort>**.



Traffic goes to Node and then to Pods.

5 LoadBalancer

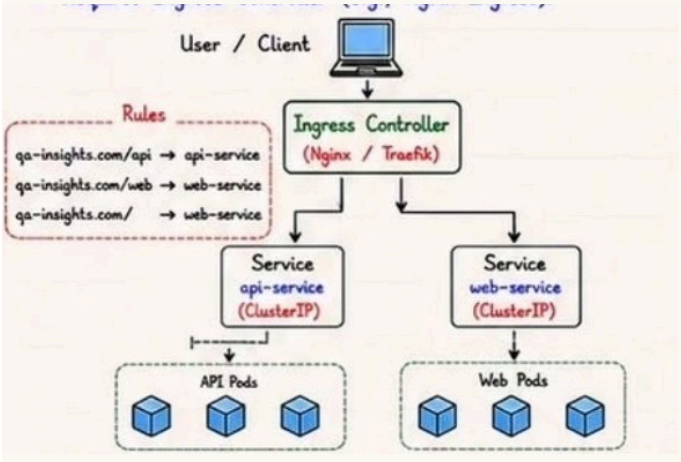
- Creates an external Load Balancer in cloud (AWS ELB, Azure LB, GCP LB).
- Provides a dedicated external IP.



Best for production applications.

6 Ingress

- HTTP/HTTPS routing to Services based on rules.
- Operates at Layer 7 (Application Layer).
- Requires Ingress Controller (e.g., Nginx Ingress).



7 Service Types Summary

Type	Exposure	Access	Use Case
ClusterIP	Cluster Internal IP	Inside Cluster	Internal services
NodePort	Node IP + Port	Inside & Outside	Dev / Testing
LoadBalancer	Cloud LB (External IP)	Outside	Production Apps
Ingress	HTTP/HTTPS (L7)	Outside (via rules)	Web Apps / APIs

💡 Remember

- ✔ Service = Stable endpoint
- ✔ Ingress = Smart traffic routing
- ✔ Use right Service type based on requirement

⭐ Interview Tip

- ✔ ClusterIP is default and not accessible from outside.
- ✔ NodePort exposes on all nodes – be careful with security.
- ✔ LoadBalancer is cloud dependent.
- ✔ Ingress = smarter routing (L7).
- ✔ Pods communicate directly using Pod IPs.

🔧 Common Commands

```
kubectl get pods -o wide
kubectl get svc
kubectl get ingress
kubectl describe svc <service-name>
kubectl describe ingress <ingress-name>
```

💡 DevOps Note

Use Network Policies to control Pod-to-Pod traffic for security. Example: Allow only QA Insights API Pods to access QA Insights DB Pods.

CHAPTER 17

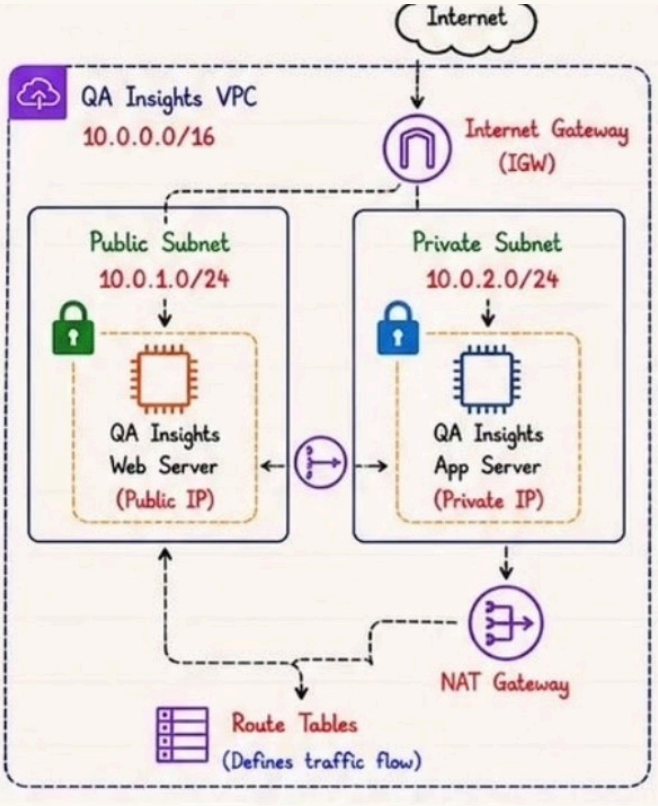
Cloud Networking (AWS VPC)

1 What is VPC?

- A Virtual Private Cloud is a logically isolated network in the cloud.
- You control IP ranges, subnets, route tables, gateways and security.
- Provides secure, scalable and flexible networking for QA Insights workloads.

> 💡 **Key Point**
> VPC is your own private network in the AWS Cloud.

2 VPC Architecture Overview



> 🔀 **Traffic Flow**
> 1. Internet → IGW
> 2. IGW → Public Subnet
> 3. Private Subnet → NAT Gateway → IGW → Internet

3 Public vs Private Subnets

Public Subnet

- ✔ Route to Internet via IGW.
- ✔ Resources have Public IP.
- ✔ Used for bastion, ALB, web servers.

Private Subnet

- ✔ No direct route to Internet.
- ✔ Resources have Private IP.
- ✔ Used for app servers, databases, caching.

4 Internet Gateway (IGW)

- Horizontally scaled, redundant, and highly available.
- Enables communication between VPC and the Internet.
- Attached to the VPC only.

Example: Internet ↔ **IGW** ↔ Public Subnet

5 NAT Gateway

- Allows instances in private subnet to access the Internet.
- Blocks inbound connections from Internet.
- Used for updates, downloads, package repositories.

Example: Private Subnet → **NAT** → **IGW** → Internet

6 Route Tables

- Defines where network traffic is going.
- Associated with subnets.
- Longest prefix match is used.
- Main Route (local): **10.0.0.0/16 → local**

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw-xxxxxxxx (for Public)
0.0.0.0/0	nat-xxxxxxxx (for Private)

7 Security Groups (Stateful Firewall)

- Operates at instance level.
- Stateful: return traffic is auto-allowed.
- Allow rules only (implicit deny all).
- Used for fine-grained control.



8 Best Practices

- ✓ Keep private resources in Private Subnets.
- ✓ Use NAT Gateway for outbound internet access.
- ✓ Use Security Groups for instance-level control.
- ✓ Use separate route tables for different subnets.
- ✓ Enable Flow Logs for monitoring traffic.
- ✓ Use least privilege access.

⭐ Interview Tip

- ✓ IGW = Internet in
- ✓ NAT Gateway = Outbound only for Private
- ✓ Route Table decides the path
- ✓ Security Group = Stateful, at instance level
- ✓ NACL = Stateless, at subnet level
- ✓ Public Subnet has route to IGW
- ✓ Private Subnet has route to NAT Gateway

💡 Common Use Cases

- ✓ Host public web apps in public subnet.
- ✓ Place application & database in private subnets.
- ✓ Allow private instances to download updates via NAT.
- ✓ Isolate environments (Dev, QA, Prod) in separate VPCs.

💡 Remember

- ✓ VPC = Your private network
- ✓ IGW = Internet access
- ✓ NAT = Outbound internet
- ✓ Route Table = Traffic path
- ✓ SG = Stateful firewall

🔧 Useful Commands (AWS CLI)

```
aws ec2 describe-vpcs            # List VPCs
aws ec2 describe-subnets          # List Subnets
aws ec2 describe-route-tables     # List Route Tables
aws ec2 describe-security-groups  # List Security Groups
aws ec2 describe-nat-gateways     # List NAT Gateways
```

CHAPTER 18

Networking Cheat Sheet

1 Key Comparisons

A. OSI vs TCP/IP

OSI Model (7 Layers)	TCP/IP Model (4 Layers)	Function
7. Application 6. Presentation 5. Session	Application	Network services to applications
4. Transport	Transport	End-to-end communication
3. Network	Internet	Logical addressing & routing
2. Data Link 1. Physical	Network Access	Framing, MAC addressing Bit transmission over media

B. TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Reliable, guaranteed delivery	Best-effort, no guarantee
Order	Maintains order of packets	No order guarantee
Speed	Slower (overhead for reliability)	Faster (low overhead)
Data Type	Stream of bytes	Datagrams (packets)
Use Cases	Web, Email, File transfer, SSH	DNS, VoIP, Gaming, Streaming
Example Ports	80, 443, 22, 3306	53, 123, 161, 500

C. HTTP vs HTTPS

Feature	HTTP	HTTPS
Full Form	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Port	80	443
Security	Not Encrypted	Encrypted (SSL/TLS)
Data	Plain Text	Encrypted
Authentication	No	Yes (Certificate)
SEO / Trust	Lower	Higher (Secure)
Use	General web traffic	Sensitive data, APIs, Logins

💡 Tip:
Use HTTPS everywhere. It encrypts data and builds trust.

2 Common Commands

Command	Purpose	Example	What it shows
ping	Test reachability and latency.	`ping qa-insights.com` `[Windows: ping qa-insights.com]`	Whether host is reachable, response time (latency) and packet loss.
traceroute	Trace the path packets take to reach destination.	`traceroute qa-insights.com` `[Windows: tracert qa-insights.com]`	All network hops (routers) and latency at each hop.
nslookup	Query DNS to get domain information.	`nslookup qa-insights.com`	IP address, Name servers and DNS records.
dig	Advanced DNS lookup (tool for admins).	`dig qa-insights.com` `dig @8.8.8.8 qa-insights.com A`	Detailed DNS records, TTL, response codes, etc.
curl	Transfer data to/from server (HTTP/HTTPS).	`curl -I https://qa-insights.com/api` `curl -X GET https://qa-insights.com/api`	HTTP headers, response code and response data.
ip	Show / configure network interfaces and routes.	`ip a` `ip route`	IP addresses, interfaces, routing table.
ss	Show socket statistics (TCP/UDP connections).	`ss -tuln` `ss -tp`	Listening ports, open connections, process info.

3 Quick Notes

Important Ports (Most Used)

Port	Service	Port	Service
22	SSH (Secure Shell)	5432	PostgreSQL
53	DNS (Domain Name System)	27017	MongoDB
80	HTTP (Web)	6379	Redis
443	HTTPS (Secure Web)	25	SMTP (Email)
3306	MySQL	21	FTP

💡 ⭐ Remember

- ✔ DNS translates domains to IPs.
- ✔ IP routes packets to destination.
- ✔ TCP = Reliable, UDP = Fast.
- ✔ Firewalls control what is allowed.
- ✔ Always use HTTPS for security.

🛡️ Best Practices

- ✔ Use least privilege access.
- ✔ Keep security groups tight.
- ✔ Monitor network traffic & logs.
- ✔ Encrypt data in transit (HTTPS/TLS).
- ✔ Document network architecture.